

FilsDesk ERP

COMPREHENSIVE FEATURE FLOW GUIDE

Complete Operational Processes, Compliance & Accounting Ledger Engine

End-to-End Enterprise Flow Documentation

A complete mapping of technical transactions and offline capabilities



Platform Architecture: Offline-First Modular Monolith (PostgreSQL + SQLite)

Designed for UAE SMEs & Federal Tax Authority (FTA) Compliance

Doc Version: 2.0 - Date: August 2026

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1. Identity, Access, & Session Security Flow

[1.1] Admin Portal User Creation

The Owner or Secondary Admin enters employee info (Name, Email, Role, and Tenant/Branch bindings) on the Access Settings page. The password field is auto-generated if left blank.

[1.2] Credentials SMTP Broadcast

The backend AuthService hooks into EmailService to send a secure HTML credentials package. If SMTP settings are offline, the credentials log automatically to the backend console simulator.

[1.3] Login Handshake

User enters credentials. The system filters by lowercase email and validates the 'isActive' flag status. If false, authentication is blocked.

[1.4] JWT Generation

Upon validation, the backend generates an asymmetric JSON Web Token containing the user's ID, Tenant context, and RBAC Role, which is returned to the client and stored in localStorage.

[1.5] RBAC Route Guard

The frontend Sidebar and Page router check localStorage mappings. Restricted pages (like Payroll or General Ledger) are completely hidden from cashier and inventory roles.

Note: Administrative accounts cannot deactivate themselves to prevent security lockouts.

2. Sales Invoicing & Ledger Posting Engine Flow

[2.1] Draft Formulation

Salesperson creates an invoice. Line items are added, calculating base rates and applying a standard 5% UAE VAT rate automatically based on the item tax profile.

[2.2] Invoice Posting Execution

The posted action triggers an immutable posting database transaction. Once posted, editing or deletion actions are disabled permanently.

[2.3] Double-Entry Posting

Only the Ledger Posting Service executes double-entry rules. It translates the invoice into balanced debits/credits in the journal register.

[2.4] Audit Constraint Check

Before committing, a database-level trigger verifies that $\text{SUM}(\text{debits}) = \text{SUM}(\text{credits})$. If they do not balance, the transaction rolls back.

[2.5] Immutable Snapshot

The invoice stores the exact tax rate snapshot used at posting time. Later configuration changes will not affect historical records.

Note: Double entry mapping: Debit Accounts Receivable (105%), Credit Sales Revenue (100%), Credit Output VAT (5%).

3. Point of Sale (POS) Billing & Hardware Integration

[3.1] POS Session Initialization

The cashier opens the POS Billing Counter (which supports barcode scanning and fast-click categories). An active Cash/Bank ledger is selected.

[3.2] Real-Time Cart Calculation

Items are added. Subtotal, discounts, and VAT are calculated on each change. Inventory levels are checked in the local cache.

[3.3] Immediate Payment Collection

Cashier records payment (Cash, Card, or Split). The system marks the POS receipt as PAID immediately.

[3.4] ESCPOS Print Command

The client compiles the receipt payload and sends it to the native Electron bridge, triggering USB/Network 80mm thermal printing.

[3.5] Cash Sales Posting

Behind the scenes, the double-entry engine posts: Debit Cash/Bank Asset, Credit Sales Revenue, Credit Output VAT.

4. Purchases, Supplier Bills, & Payables Flow

[4.1] Purchase Bill Registration

The accountant registers a supplier bill (specifying Supplier TRN, supply date, item units, and purchase rates).

[4.2] Input VAT Recovery

The system extracts the recoverable 5% Input VAT. Items without valid supplier TRNs are classified as out-of-scope.

[4.3] Inventory Stock-In Trigger

Posting the purchase bill automatically fires an inventory movement event to increase physical stock levels at the designated branch/warehouse.

[4.4] Supplier Payables Ledger

Double entry: Debit Inventory Asset or Expense, Debit Recoverable Input VAT, Credit Accounts Payable (Supplier Ledger).

[4.5] Supplier Reconciliation

When payments are disbursed, a payment record is generated: Debit Accounts Payable, Credit Bank. Payment is matched to the specific purchase bill.

5. Inventory Control, Warehouses, & Movements

[5.1] Stock Intake

Stock quantities are updated either manually via adjustments or automatically via finalized supplier purchase bills.

[5.2] Inter-Depot Transfers

An authorized inventory manager issues a Transfer Order: Source Depot -> Destination Depot. Stock is decremented from source and incremented at destination in a single transaction.

[5.3] Damage / Shrinkage Adjustments

Damage loss entries are recorded. The inventory value is written off, posting: Debit Inventory Adjustment Expense, Credit Inventory Asset.

[5.4] Cost of Goods Sold (COGS)

Upon sales invoice posting, the inventory engine computes the cost of sold stock using the configured method (Weighted Average or FIFO) and posts COGS updates to the general ledger.

Note: Direct cross-module writing to inventory tables is disabled. All adjustments must go through service APIs.

6. Employee Records, Payroll, & Payslips Delivery

[6.1] Employee Master Registry

Employee details, department, basic salary, allowances, and monthly deduction variables are maintained under the payroll system.

[6.2] Payroll Run Execution

Each month, the payroll system computes gross salaries (Basic + Allowances) and net payouts (Gross - Deductions).

[6.3] Payslip Generation

The system generates a stylized, premium HTML payslip matching the corporate format.

[6.4] Secure Email Dispatch

The Payslip delivery service connects to EmailService to broadcast the payslip PDF/HTML directly to the employee's email address.

[6.5] Ledger Payroll Accruals

Double entry: Debit Salary Expense, Credit Wages Payable. On payout: Debit Wages Payable, Credit Bank.

7. UAE VAT 201 Tax Return Mapping & Compliance

[7.1] Compliance Configuration

Tax rules (standard rate 5%, zero-rated, exempt) are configured globally. Valid TRN formats and rules conform to Federal Tax Authority guidelines.

[7.2] Tax Transaction Log

Every posted invoice and purchase bill updates a centralized tax ledger table which holds the legal supply date, base currency AED total, and tax values.

[7.3] VAT 201 Return Compiler

The reporting engine compiles standard boxes: Box 1a (Sales & Output VAT), Box 10 (Recoverable Purchases & Input VAT), and Net Payable.

[7.4] Immutable Returns Period Close

Once the tax period is filed, the period is locked. No past transactions can be posted into this period.

Note: Golden rule: UAE VAT return mappings are generated strictly from posted ledger data, never drafts.

8. Offline-First Sync Architecture

[8.1] Offline Transaction Commit

The client (mobile or desktop POS) creates a transaction offline. The record is written to a local SQLite database (using Drift).

[8.2] Sync Outbox Queue

The local SQLite database appends a record in the local outbox sync queue containing the request payload and a client-generated UUID.

[8.3] Background Sync Daemon

A background client daemon monitors internet connectivity. When online, it POSTs outbox mutations using client-generated Idempotency Keys.

[8.4] Server Ingestion & Conflict Resolution

The backend Sync Module verifies the payload, permissions, and row versions. If conflicts are resolved, it commits to PostgreSQL.

[8.5] Cursor Progress Advance

Upon successful remote commit, the server returns the updated state. The client updates its local outbox and advances its sync cursor.

Note: Every client mutation carries a unique UUID, device ID, and client timestamp to avoid duplicate posting on server retries.